BRANCH C · ANALYTICS STUDY

Better analytics: options

Chart ideas for the app, built from the statistics study and drawn in the app's own style. Every chart here is real math run on the 1-year sample data, not made-up numbers. Nothing is built yet. Branch B (feedback) and main are untouched.

What's inside

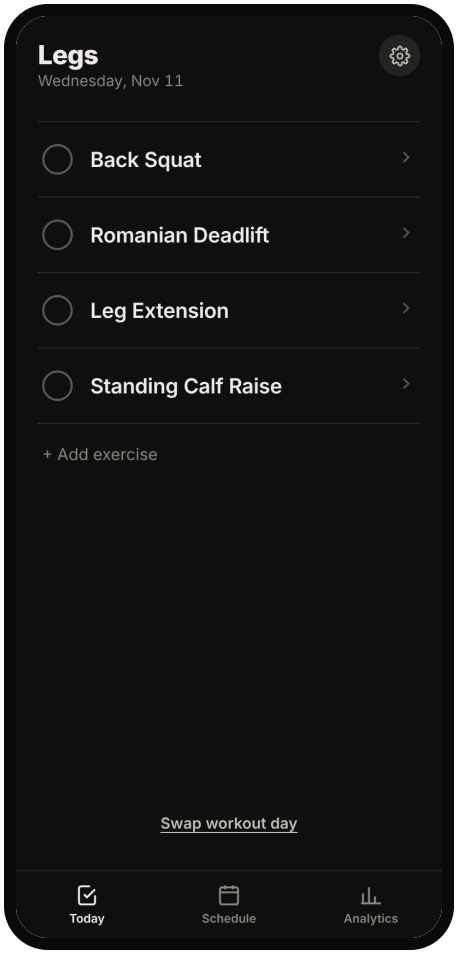
1. Where things live today (current screens)
2. End-of-workout charts: 5 options for set-to-set rep drop-off
3. New Analytics cards, each tied to a stats idea
4. A short pick list so you can circle what you want

How to read each card

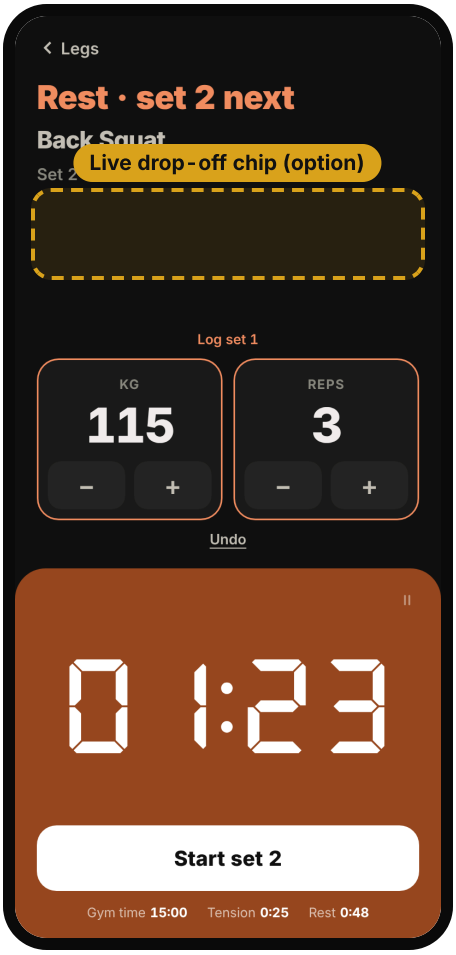
STATS IDEA which concept from the study it uses WHERE where it would live in the app.
The dark card is what it would look like on your phone. The note under it says what it tells you, in plain words.

Where things live today

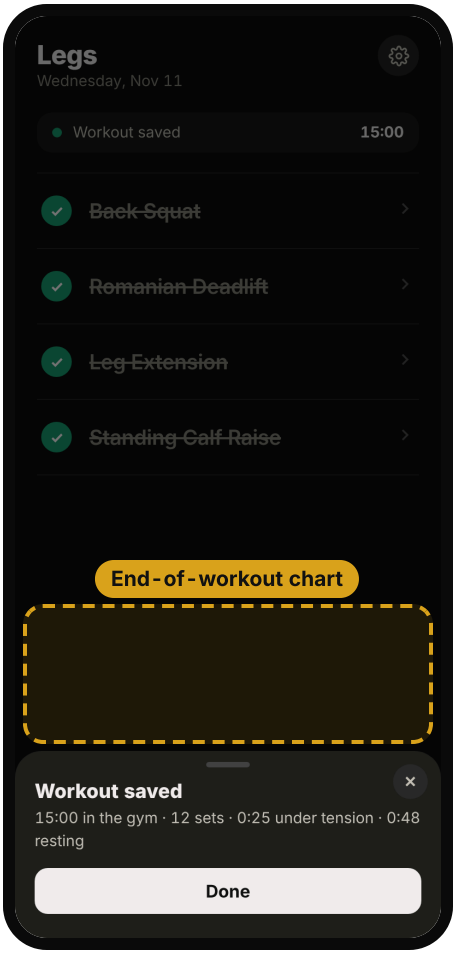
Yellow boxes mark where the new charts would go.



Today
The checklist. No charts here, on purpose.



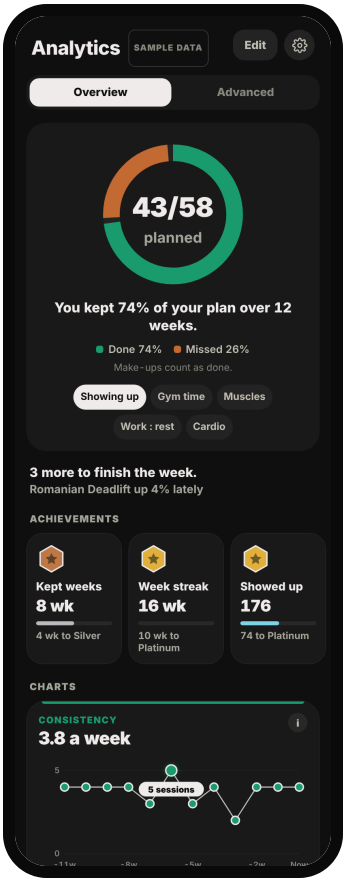
During the workout
Half list, half timer. Keep it clean.



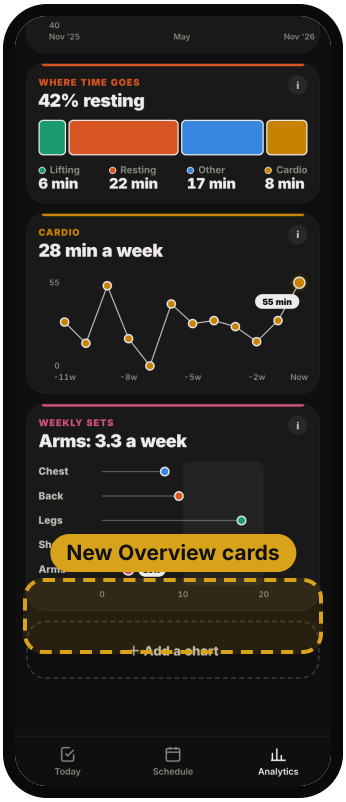
Workout saved
Today it's one line of text. This is the spot for the daily chart.

Analytics today

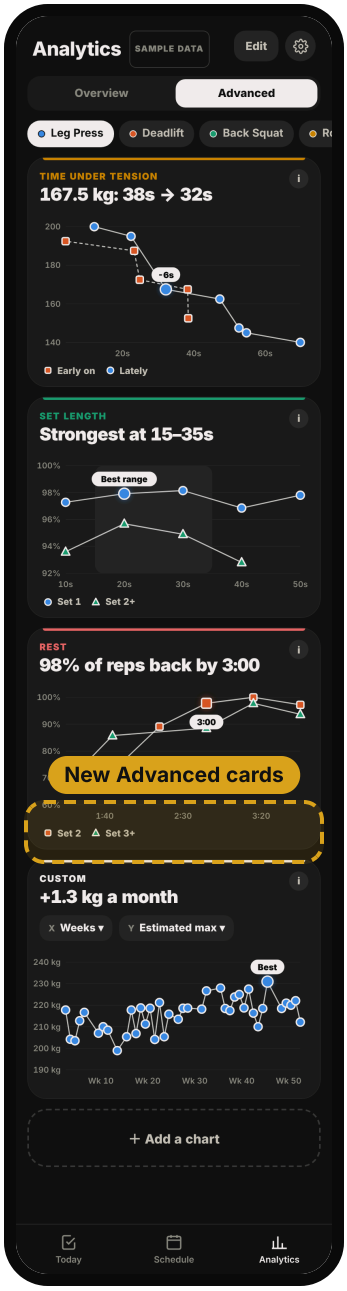
Overview (top and bottom of the scroll) and Advanced, as they are on branch B. New cards would slot into these lists, in the same card style.



Overview, top



Overview, bottom



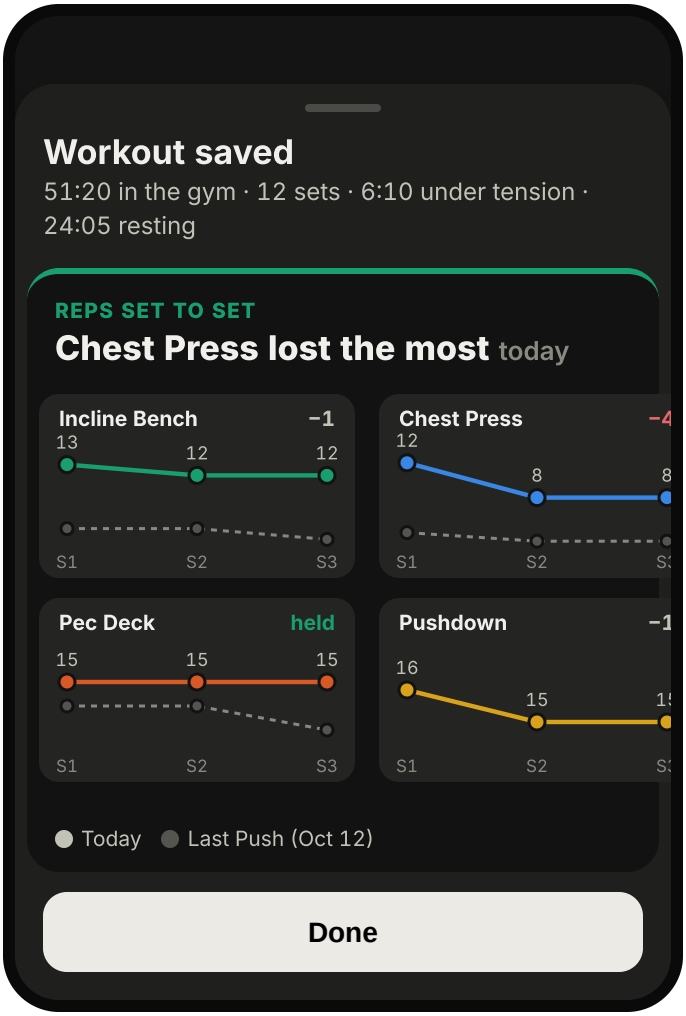
Advanced

The Advanced tab already uses some of the stats: **Rest** (98% of reps back by 3:00) is close to a regression, and **Set length** is a grouped average. The new cards build on these rather than replace them.

END OF WORKOUT

Set-to-set rep drop-off: 5 options

Each shown inside the Workout saved sheet, using a real Push day from the sample (Mon Nov 2). Compared with the Push before it.

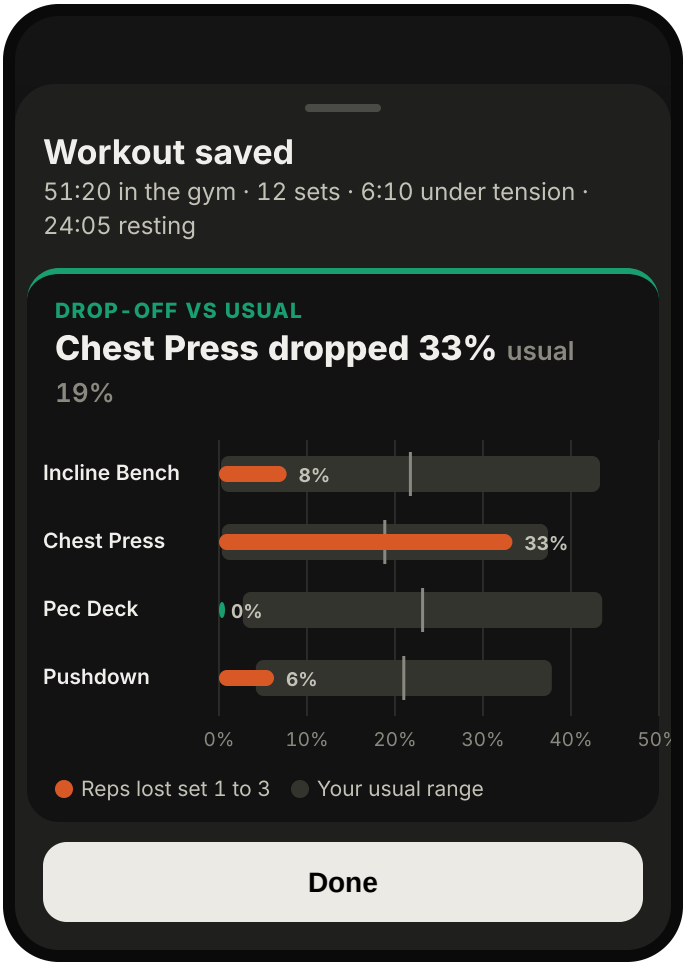


A. Small lines per exercise

One mini chart per exercise: today's reps across sets in color, last time as a faint dashed line. The number top right is reps lost from set 1 to the last set.

- Most direct answer to "how did my reps fall off?"
- Easy to compare with last time at a glance
- Grows tall with 5 to 6 exercises

DESCRIPTIVE TIME SERIES (TINY)

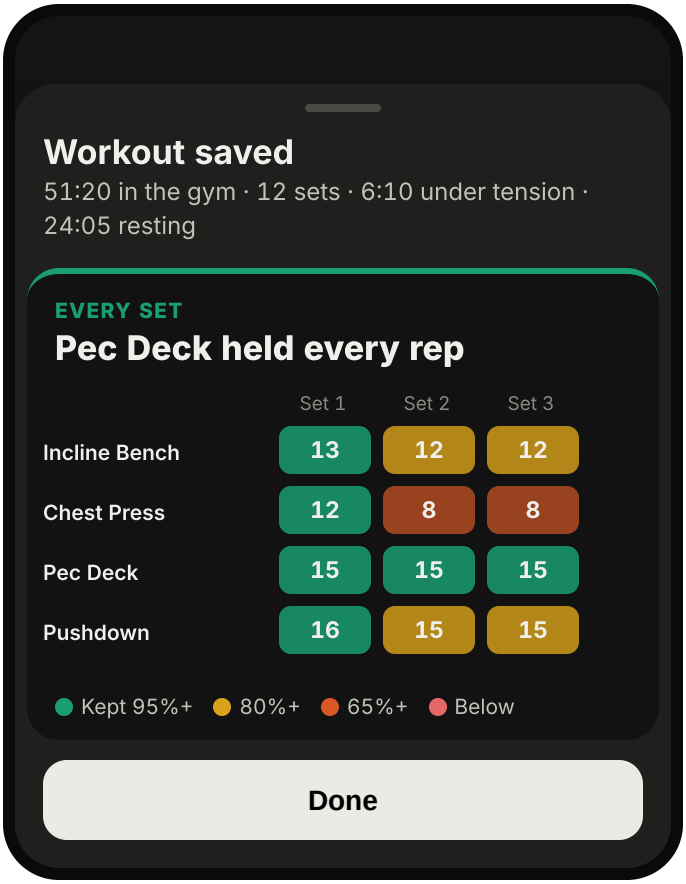


B. Drop-off bars against your usual range

One bar per exercise: % of reps lost from set 1 to set 3. The gray band is your normal range (average, plus or minus one standard deviation from your history), with a line at your average.

- Tells you if today was normal, not just what happened
- Red = more fatigue than usual, green = held up better
- Stays short no matter how many sets

MEAN STANDARD DEVIATION

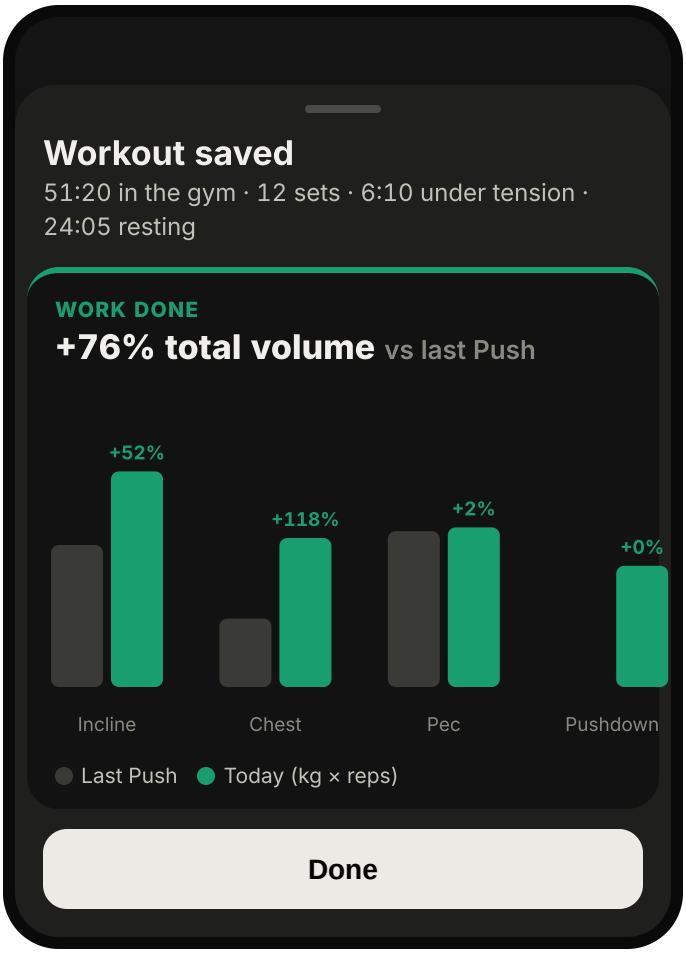


C. Set grid

A tiny table: exercises down, sets across. Each square shows the reps, colored by how much of set 1 you kept.

- The most compact, reads like a scoreboard
- Shows the exact rep counts
- Less visual than a line

DESCRIPTIVE HEAT MAP



D. Volume today vs last time

Weight × reps per exercise, today next to last time. Percent change on top.

- One number that blends weight and reps
- Doesn't show the set-to-set fall itself
- Good as a second chart under A or B

BAR PLOT



E. Rest vs reps kept

Every set after the first is a dot: how long you rested, and what % of set 1 reps you got. Circles are set 2, triangles set 3, colored by exercise.

- Explains why a drop happened
- Links straight to the rest timer, the heart of the app
- Few dots on a single day, so better as a weekly view

LINEAR REGRESSION (VISUAL)

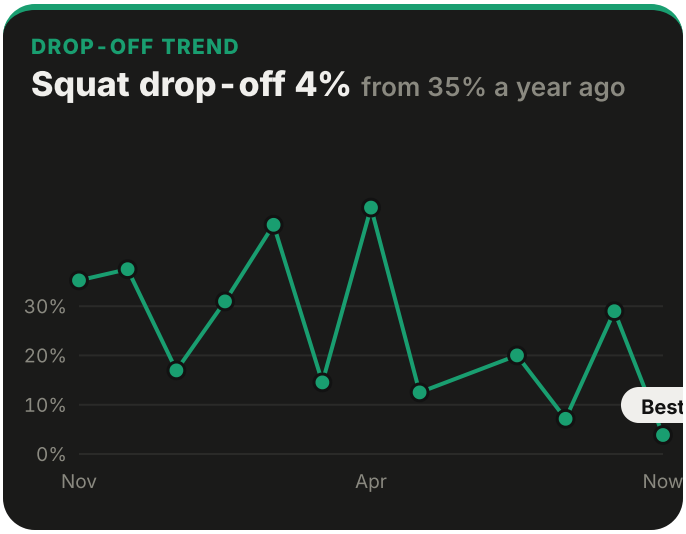
My pick: B in the sheet (it answers "was today normal?"), with a tap to open A for the detail. That keeps the sheet short, so it never feels like homework after a workout.

ANALYTICS TAB

New cards, one per stats idea

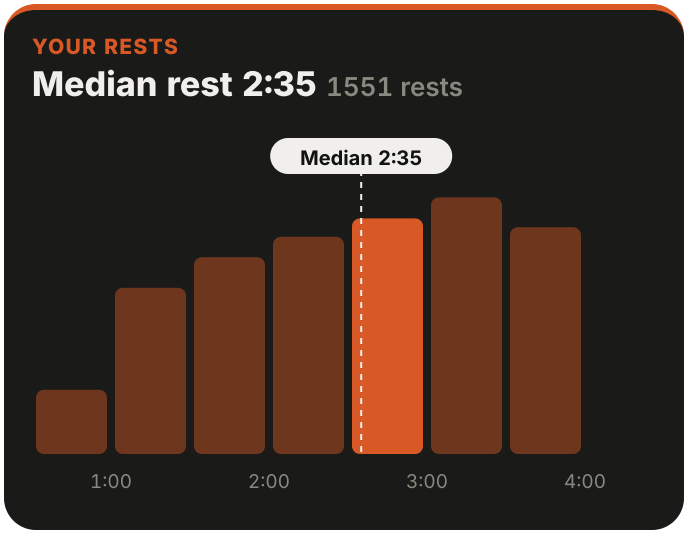
All on the sample year. Squat is used where one exercise is needed; each card would have the same exercise picker the app already has.

TIME SERIES · ADVANCED



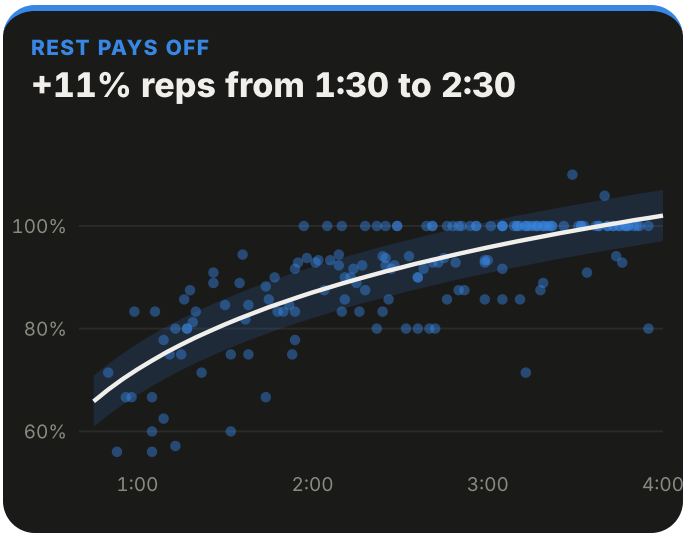
Average % of reps lost from set 1 to set 3, month by month. Going down means your conditioning between sets is improving, even if the weight on the bar isn't moving.

HISTOGRAM · MEDIAN · OVERVIEW



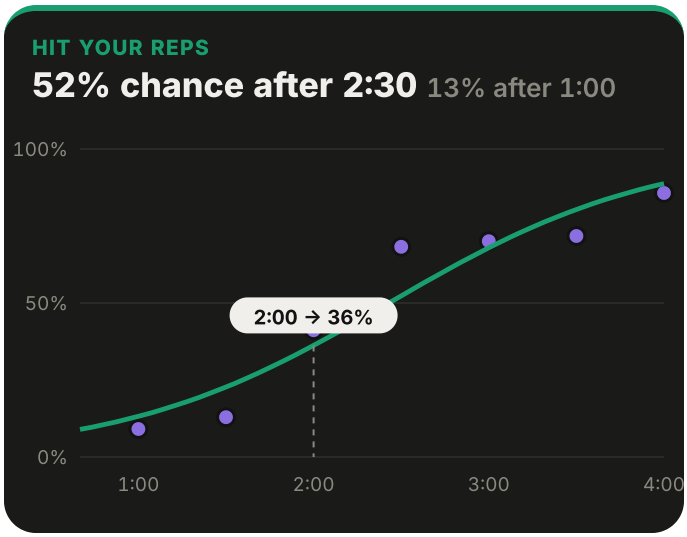
How long you actually rest, as a histogram. The median ignores the odd 6-minute chat. Pairs with the rest timer: is your target realistic?

REGRESSION · ADVANCED



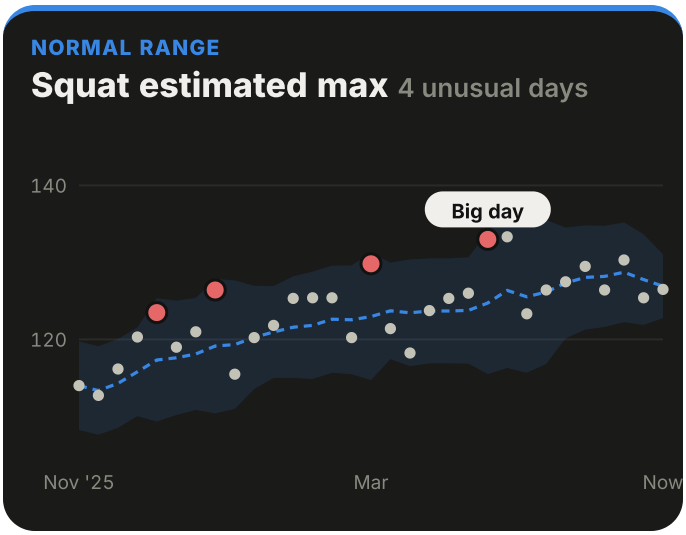
Every set 2 from the year (587 sets): rest on the bottom, % of set 1 reps on the side, with the best-fit curve and its spread. Tells you how much extra rest is actually worth.

LOGISTIC REGRESSION · ADVANCED



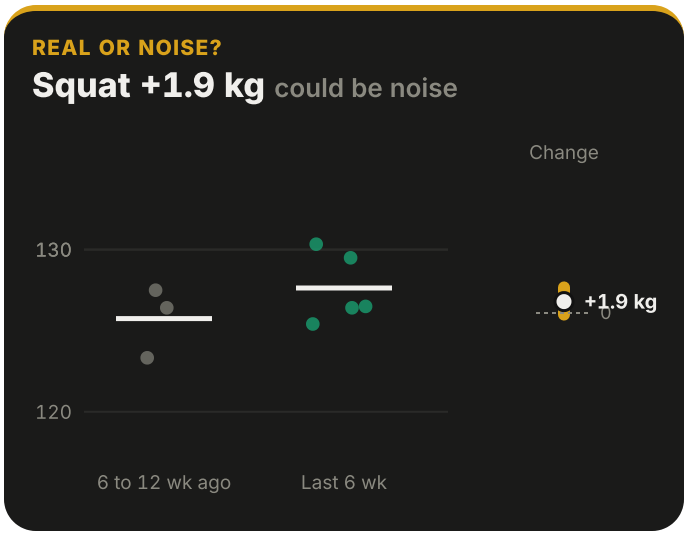
The chance set 3 lands within 1 rep of set 1, by how long you rested. Could later set the rest timer for you: "rest 2:15 for a 70% chance".

STD DEVIATION · ANOMALY · ADVANCED



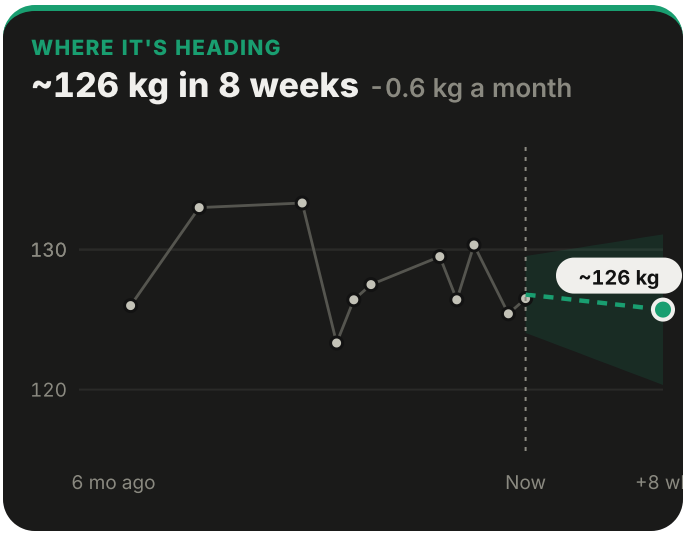
Each dot is a session's estimated max. The band is your recent normal (average ± 2 standard deviations). Red dots fall outside it: an off day worth a note (sleep? short rest?) or a real jump.

HYPOTHESIS TEST · RESAMPLING · OVERVIEW



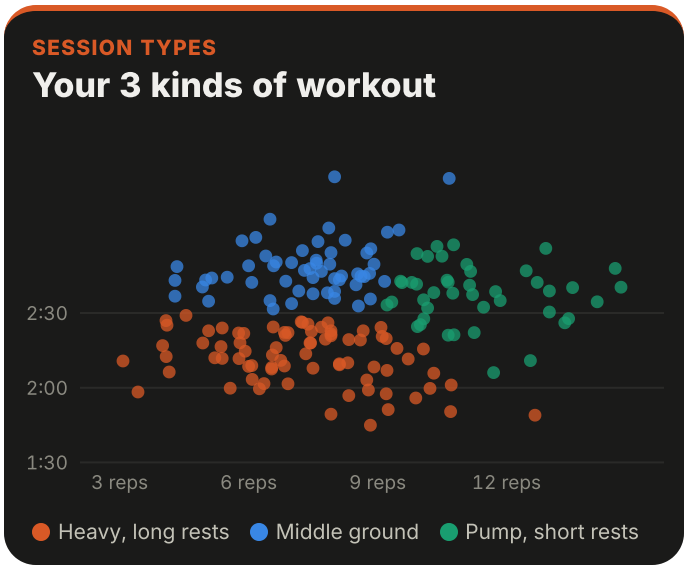
Last 6 weeks vs the 6 before. The bar on the right is the likely range of the true change, from reshuffling your sessions 3,000 times. If it sits above 0, the gain is real. Here the range is -0.3 to +4.3 kg.

TIME SERIES · LINEAR REGRESSION · OVERVIEW



A straight-line trend from the last 4 months, carried forward 8 weeks. The band widens because the future is less certain. Good for setting a realistic target.

K-MEANS CLUSTERING · ADVANCED



Every session placed by average reps and average rest, then grouped automatically. Heavy, long rests 43%, Middle ground 31%, Pump, short rests 24%. Shows how your training actually splits, without you labeling anything.

Pick list

Circle what you want and I'll build those on branch C. Everything is based on data the app already records, so there's no extra logging in the gym.

CHART	WHERE	STATS IDEA	WHAT IT ANSWERS	EFFORT
B. Drop-off vs usual	End of workout	Mean · SD	Was today's fatigue normal?	Small
A. Lines per exercise	End of workout	Descriptive	How did reps fall set to set?	Small
C. Set grid	End of workout	Descriptive	Exact reps, at a glance	Small
D. Volume vs last time	End of workout	Bar plot	Did I do more work?	Small
E. Rest vs reps kept	End of workout	Regression	Did short rests cost me?	Small
Drop-off trend	Advanced	Time series	Is my set-to-set endurance improving?	Small
Your rests	Overview	Histogram · Median	How long do I really rest?	Small
Rest pays off	Advanced	Regression	How much is extra rest worth?	Medium
Hit your reps	Advanced	Logistic regression	How long to rest to hit my reps?	Medium
Normal range	Advanced	SD · Anomaly	Was that an off day or a real change?	Medium
Real or noise?	Overview	Hypothesis test · Resampling	Am I actually getting stronger?	Medium
Where it's heading	Overview	Time series · Regression	Where will I be in 2 months?	Medium
Session types	Advanced	K-means	What kinds of workouts do I really do?	Larger

Notes

Charts need enough history to mean anything. Like the existing cards, each would stay hidden or show "a few more sessions" until there's enough of your real data.

The live drop-off chip during the workout is an option, but it adds something to the clean timer screen, so I'd leave it out unless you want it.